

Evaluation: Metrics and Benchmarking

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Evaluation in NLP and Large Language Models

What is Evaluation?

Assessing how well a model, system, or algorithm performs on a given task.

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- Verify the quality of model outputs
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How Do We Evaluate NLP Systems?

- **Automatic metrics** (BLEU, ROUGE, BERTScore)
- **Human evaluation** (fluency, adequacy, faithfulness)
- **Benchmarks and LLM judges**

How Do We Evaluate Quality?



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Question: Is this a good cake?

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Human Evaluation

- Taste
- Texture
- Appearance
- Freshness

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Automatic Measurements

- Sugar level
- Calories
- Moisture
- Weight

Insight

No single metric captures overall quality.

Why Evaluation Matters?

Core Question

If a Large Language Model generates an answer, how do we know if it is **good**?

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In traditional ML tasks, evaluation is straightforward:

Task	Evaluation Metric
Image Classification	Accuracy
Speech Recognition	Word Error Rate (WER)
Machine Translation	BLEU
Large Language Models	???

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- Generative models can produce **many valid outputs**.
- Surface similarity does not always reflect **semantic correctness**.
- Models may generate **hallucinated information**.

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Evaluators judge system outputs based on:

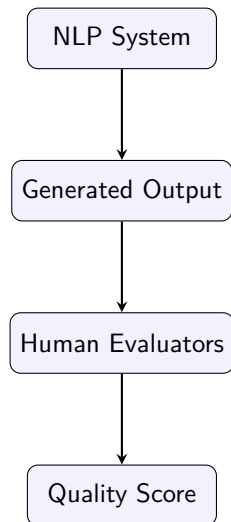
- **Fluency** – Is the output grammatically correct?
- **Adequacy** – Does it preserve the meaning?
- **Faithfulness** – Is it factually grounded?

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Machine Translation Example

Source (French): *Le chat est sur le tapis*

Reference Translation

- The cat is on the mat.

Candidate Translations

- Candidate 1: The cat is near the mat.
- Candidate 2: Cat is on mat the.
- Candidate 3: The animal is on the rug.
- Candidate 4: The cat is on the carpet.

Which candidate is the best translation?

Issues in Translation Candidates

Reference: The cat is on the mat.

Candidate	Translation	Issue
Candidate 1	The cat is near the mat	Low Adequacy (meaning changed)
Candidate 2	Cat is on mat the	Low Fluency (ungrammatical)
Candidate 3	The animal is on the rug	Low Fidelity (semantic drift)
Candidate 4	The cat is on the carpet	Correct meaning but less precise (moderate fidelity)

Why Automatic Evaluation Metrics?

Human Evaluation

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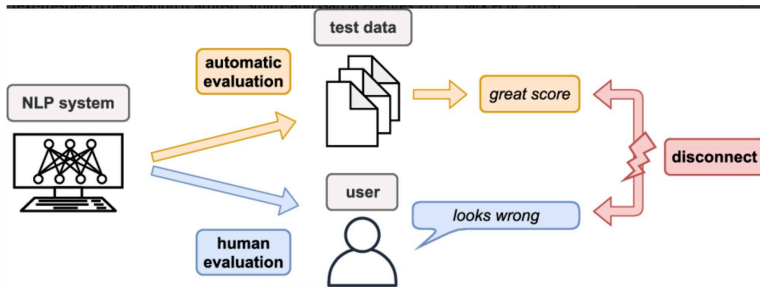
The Bottleneck

- **Expensive:** High labor costs.
- **Slow:** Takes weeks/months.
- **Inconsistent:** Hard to reproduce.
- Subjective judgments



Does Automatic Evaluation Solve the Issue?

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- Automatic metrics evaluate model outputs on **test data**.

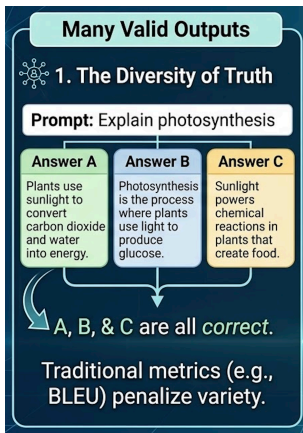
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Many Valid Outputs



1. The Diversity of Truth

Prompt: Explain photosynthesis

Answer A

Plants use sunlight to convert carbon dioxide and water into energy.

Answer B

Photosynthesis is the process where plants use light to produce glucose.

Answer C

Sunlight powers chemical reactions in plants that create food.

A, B, & C are all *correct*.

Traditional metrics (e.g., BLEU) penalize variety.

Surface vs. Semantics



2. Shallow Similarity vs. Deep Meaning

Example 1: Reference vs. A

Reference The medication **is** safe.

Model A The medication **is** safe.



High overlap
vs. Correctness

Example 2: Reference vs. B

Reference The medication **is** safe.

Model B The medication is **not** safe.



Still very high
Correctness

A single word (e.g., **NOT**) reverses the entire semantic meaning, yet traditional metrics can give a falsely high score.

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Hallucinated Information



3. Confident & Fluent Hallucinations

Source Document

PRODUCT MANUAL
(Ground Truth)

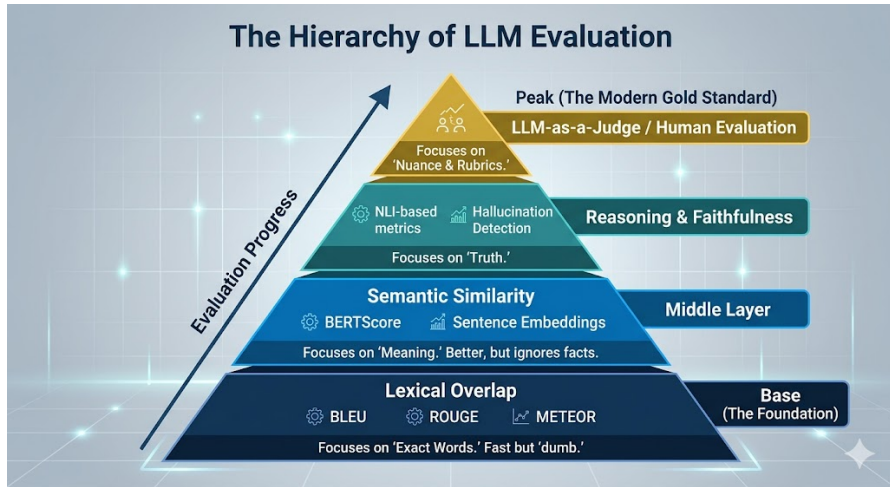
...Warranty covers **mechanical** failures only.



CONFIDENT CLAIM:
"We cover **all** **cosmetic** damage!"

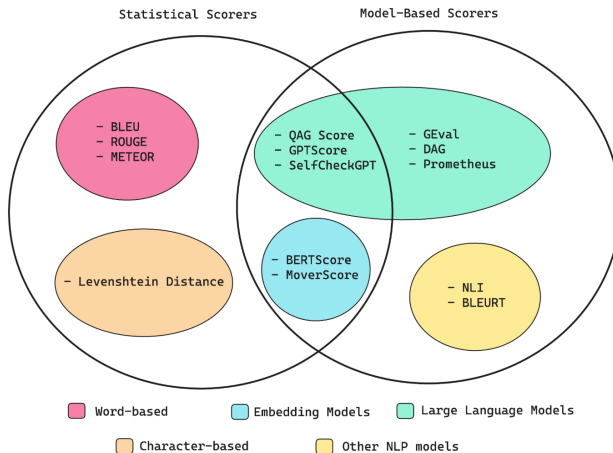
The model is 100% fluent, but 100% **false**.
Classic metrics (e.g., ROUGE) see fluency but ignore truthfulness.

Hierarchy of LLM Evaluation



- Evaluation isn't a single checkmark
- it's a progression from lexical accuracy to semantic truth and human-aligned nuance.

The Evaluation Metrics



- **The Rise of Automatic Metrics:** To accelerate research, n-gram overlap metrics were introduced:
 - **BLEU – Bilingual Evaluation Understudy**
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- **Legacy:** These metrics became the **industry gold standard** for nearly two decades (2002–2022), defining the "Leaderboard Era" of NLP.

References:

- Papineni, K., et al. (2002). *BLEU: a Method for Automatic Evaluation of Machine Translation*. ACL.
- Lin, C. Y. (2004). *ROUGE: A Package for Automatic Evaluation of Summaries*. ACL Workshop.

BLEU: Bilingual Evaluation Understudy

Idea: Measure how many **n-grams** in the candidate translation appear in the reference translation.

BLEU Score

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

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c = candidate length r = reference length

BLEU : N-gram Precision

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N-gram	Candidate N-grams	Matches in Reference	Score
1-gram	The, cat, sits, on, mat	The, cat, on, mat	4/5
2-gram	The cat, cat sits, sits on, on mat	The cat	1/4
3-gram	The cat sits, cat sits on, sits on mat	None	0/3
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BLEU combines these scores using **geometric mean** and applies a **brevity penalty** for short translations.

ROUGE: Recall-Oriented Understudy for Gisting Evaluation

Idea: Measure how much of the **reference content** appears in the generated text.

ROUGE-N (n-gram recall)

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ROUGE-L (Longest Common Subsequence)

$$Recall = \frac{LCS(candidate, reference)}{\text{length}(reference)}$$

$$Precision = \frac{LCS(candidate, reference)}{\text{length}(candidate)}$$

Example

Reference: *The cat is on the mat*

Candidate: *Cat on mat*

- Unigram matches: cat, on, mat
- ROUGE-1 Recall = $3/6$

Key Idea: ROUGE focuses on **coverage of reference content**, widely used for **text summarization**.

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- Stem matches (run → running)
- Synonym matches (car → automobile)
- Word order penalty

Levenshtein Distance: Edit-Based Evaluation

Idea

Measure the **minimum number of edits** required to transform one string into another.

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Applications

- Speech recognition evaluation, OCR systems, Spelling correction

Beyond N-grams: The Need for Semantic Metrics

- **The Paraphrasing Problem:** Traditional metrics like BLEU and ROUGE rely on exact surface-level word matching. They heavily penalize perfectly valid paraphrases.

Beyond N-grams: The Need for Semantic Metrics

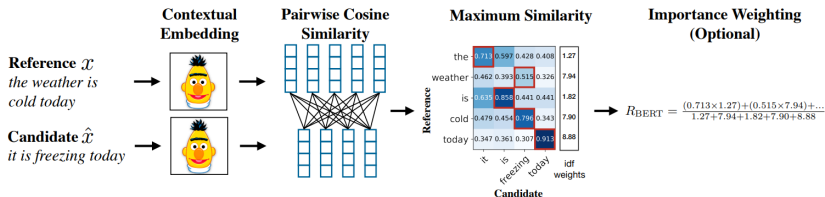
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- **Example:**
 - *Reference:* "The tiny feline rested on the rug."
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 - **BLEU Score:** 0 (No overlapping n-grams, despite identical meaning!)

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- **Example:**
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 - **BLEU Score:** 0 (No overlapping n-grams, despite identical meaning!)
- **The Deep Learning Solution:** Instead of matching words, what if we match the *meaning* of the words? We can use dense vector representations (Embeddings) from Transformer models.

BERTScore: Token-Level Semantic Overlap

- **How it works:** Uses contextualized embeddings (like BERT). For each token in the reference x and candidate $\hat{x}$, it computes the cosine similarity.
- **Mathematical Formulation:** It uses greedy matching to find the highest similarity score for each token.



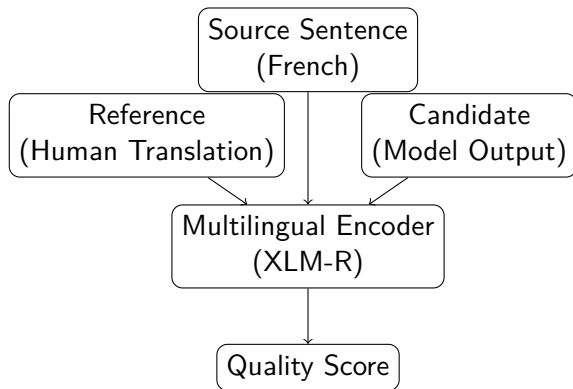
- **Advantage:** Computes Precision and Recall based on *semantic* meaning, naturally handling synonyms and paraphrasing without penalizing the model.

COMET: Neural Evaluation for Machine Translation

COMET (Rei et al., 2020) is a **neural evaluation metric** for machine translation.

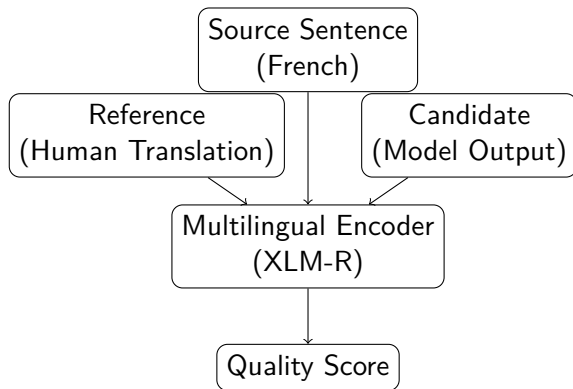
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- Uses **semantic representations** instead of surface n-gram overlap.
- Considers **source, reference, and candidate** simultaneously.
- Better correlation with **human judgments**.

Summary: The Trade-offs of Semantic Metrics

- **The Good:**

- Correlate much better with human judgments than BLEU/ROUGE.
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- **Opaque / Black Box:** They give you a score (e.g., 0.85), but they cannot explain *why* the sentence is bad.
- **Requires a Reference:** These metrics still require a human-written "gold standard" reference to compare against. They cannot evaluate open-ended chat or creative writing where no reference exists.

Evolution of Automatic Evaluation Metrics

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- reasoning
- factual correctness
- helpfulness
- open-ended responses

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Next Step: Using Large Language Models as Judges

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- But **slow, expensive, and unscalable**

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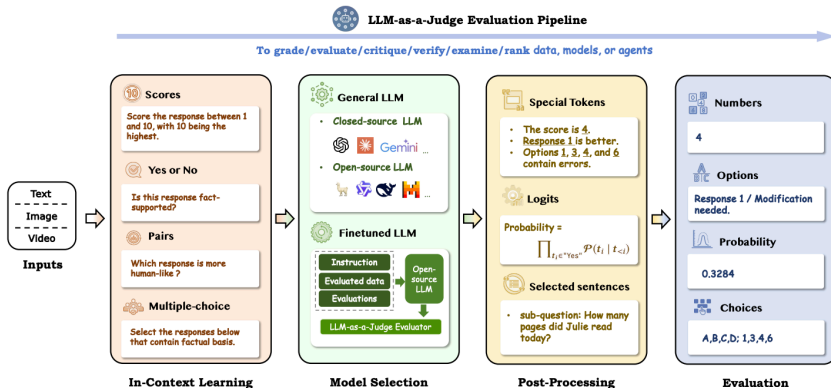
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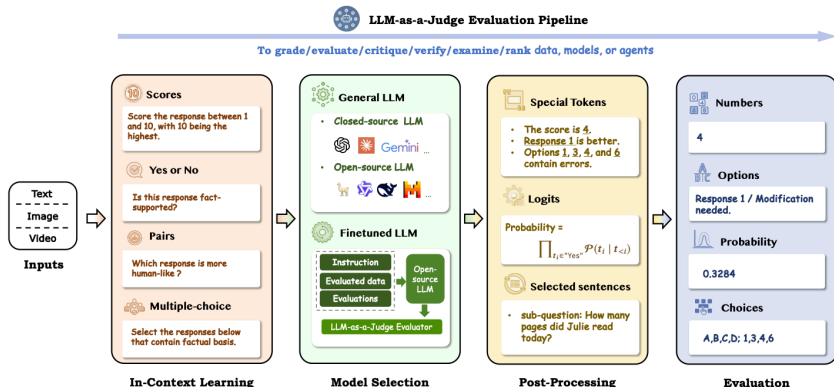
Idea

Use strong LLMs to automatically evaluate model outputs.

LLM-as-a-Judge Evaluation Pipeline



LLM-as-a-Judge Evaluation Pipeline

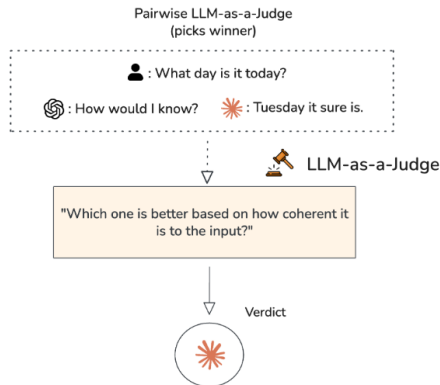


- LLMs can evaluate outputs by **scoring, ranking, or comparing responses**.
- The pipeline includes **prompt design, model selection, post-processing, and final evaluation**.

Reference: <https://arxiv.org/pdf/2411.15594>

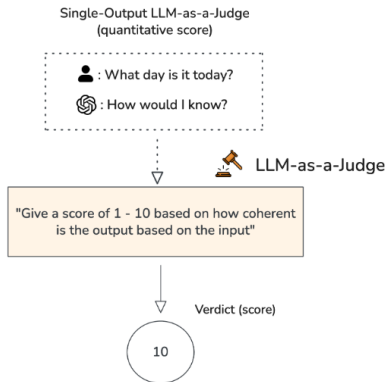
Types of LLM Judging: Pairwise Comparison

- **Pairwise Comparison:** The judge is shown Answer A and Answer B, and asked to declare a winner or a tie. (Mirrors how humans naturally compare things).



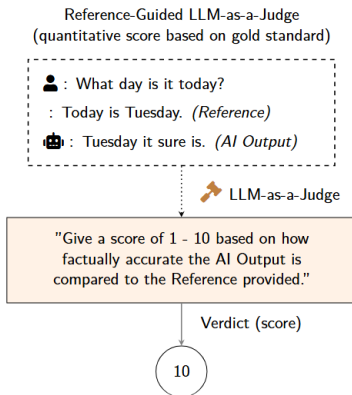
Types of LLM Judging: Single Answer Grading

- **Single Answer Grading:** The judge is asked to score a single response on a scale (e.g., 1 to 10) based on helpfulness and relevance.



Types of LLM Judging: Reference-Guided Grading

- **Reference-Guided Grading:** The judge is given the "correct" answer as a reference and evaluates how well the model's answer matches it (best for math or factual questions).



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- This makes LLM judges a **scalable proxy** for human evaluation

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LLM judges can introduce systematic biases.

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- Example:
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- **Verbosity Bias**

- Longer responses are often rated higher.
- Example:
 - A: Paris is the capital of France.
 - B: Paris, the vibrant capital city of France, is known for its culture and history.

The judge may prefer B even though A is perfectly correct.

Self-Preference Bias

- Judges may favor outputs from models similar to themselves.
- Example:
 - Output from GPT-style model
 - Output from a smaller model

The judge may prefer responses written in its own style.

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- LLM judges can only evaluate tasks they understand
- Difficult domains
 - advanced mathematics
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Takeaway

LLM judges are powerful but require careful prompt design and validation.

Hallucination in Large Language Models

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Problem

Users may trust hallucinated answers because they sound confident.

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- **Fabrication**

- Completely invented entities or citations
- Example: fake research papers

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 - Knowledge graphs
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- **Uncertainty Estimation**

- Entropy
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Benchmarking Large Language Models

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A standardized dataset used to evaluate and compare models.

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- **Goodhart's Law**

- When a metric becomes a target, it stops being a good metric

Idea

Instead of fixed datasets, evaluate models using **live human interactions**.

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Example: **LMSYS Chatbot Arena**

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- Ranking computed using Elo rating

Open Problem 1: Evaluating Reasoning

Current metrics struggle to evaluate **reasoning quality**.

Example

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The train travels 180 km because $60 \times 3 = 180$.

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The train moves continuously and therefore covers a large distance.

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Both answers may appear fluent, but only one has correct reasoning.

Research Question

How can we automatically evaluate reasoning steps?

Open Problem 2: Detecting Hallucinated Knowledge

LLMs often produce confident but incorrect facts.

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- Prompt: Who invented the telephone?

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- Reality:
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Research Question

How can systems detect hallucinated facts automatically?

Possible directions:

- knowledge graph verification
- retrieval-based fact checking
- uncertainty estimation

Open Problem 3: Multilingual Evaluation

Most evaluation benchmarks are designed for **English**.

Example

- English riddle:
What has keys but cannot open locks?

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Models often fail reasoning in other languages.

Research Question

How can we design fair multilingual evaluation benchmarks?

Open Problem 4: Evaluating Dialogue Systems

Conversation quality is difficult to measure automatically.

Example

User: I feel stressed about my exams.

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Response A: Study harder and you will succeed.

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User: I feel stressed about my exams.

Response A: Study harder and you will succeed.

Response B: That sounds stressful. Do you want help planning your study schedule?

Both responses are fluent, but B is more **empathetic**.

Other Research Question

How can evaluation capture empathy and helpfulness?

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